



Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic business decisions.

Dataset at a Glance

3,900

Total Purchases

Transactional records across all product categories

18

Columns

Demographics, purchase details & shopping behavior

37

Missing Values

Only in Review Rating — imputed by category median

\$59.76

Avg Purchase

Mean purchase amount across all transactions

Key features include customer demographics (Age, Gender, Location, Subscription Status), purchase details (Item, Category, Amount, Season, Size, Color), and shopping behavior (Discount Applied, Previous Purchases, Frequency, Review Rating, Shipping Type).

Data Preparation in Python

01

Load & Explore

Imported via pandas; used `df.info()` and `.describe()` for structure and statistics.

02

Clean & Standardize

Imputed missing Review Ratings by category median; renamed columns to snake_case.

03

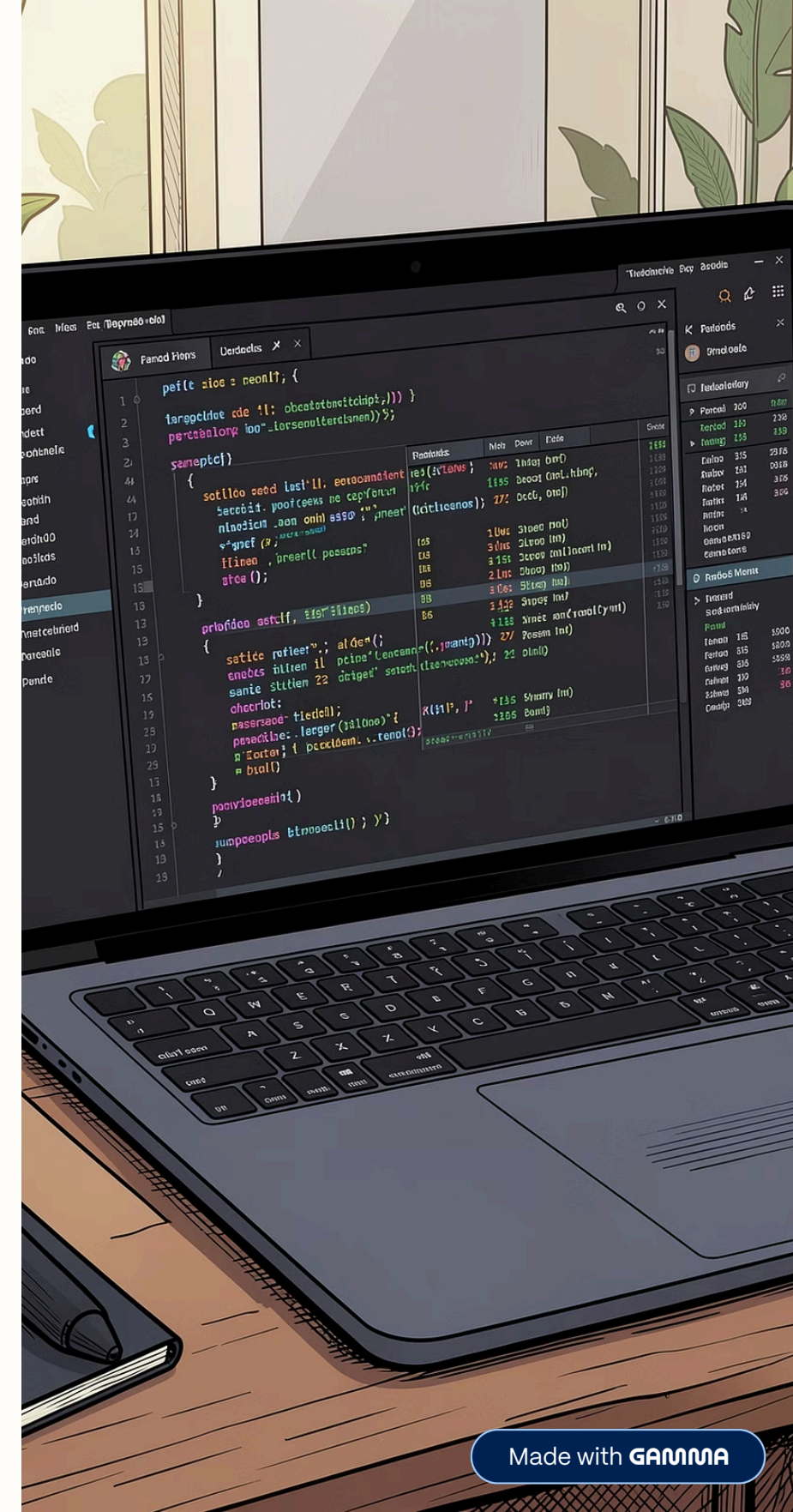
Feature Engineering

Created `age_group` bins and `purchase_frequency_days`; dropped redundant `promo_code_used`.

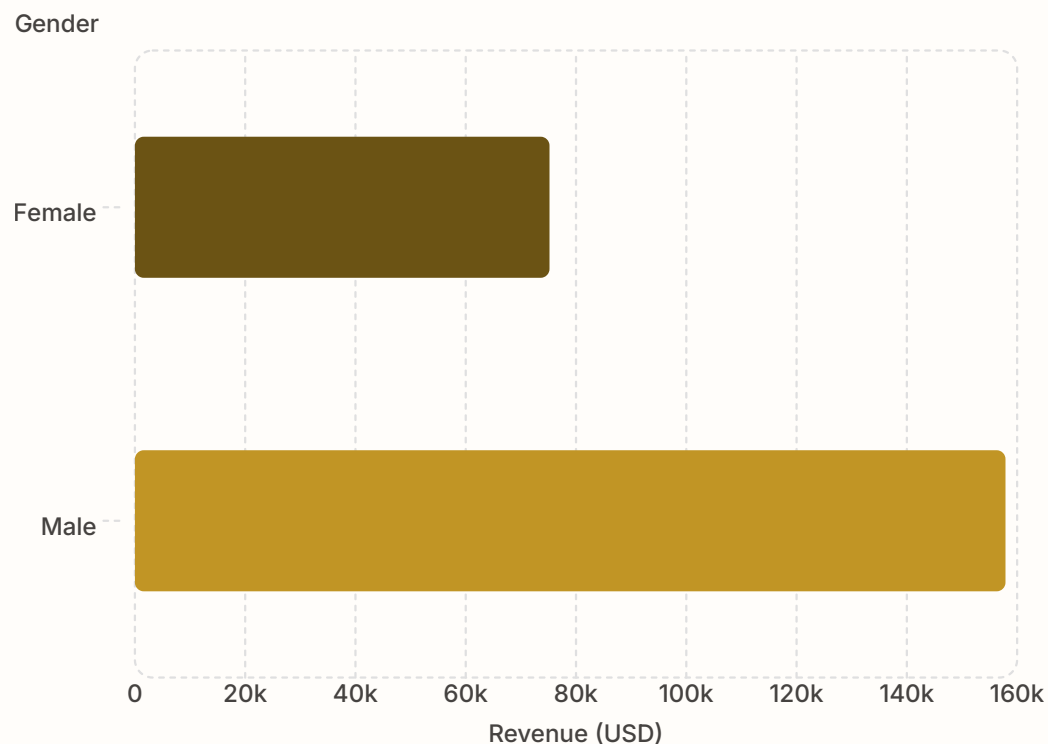
04

Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis.



Revenue by Gender



Male Customers Drive Revenue

Male customers generated **\$157,890** in total revenue — more than double the **\$75,191** from female customers. Of 3,900 records, 2,652 were male purchases.

- This gap warrants investigation into product mix, marketing reach, and purchasing frequency by gender.

Subscribers vs. Non-Subscribers



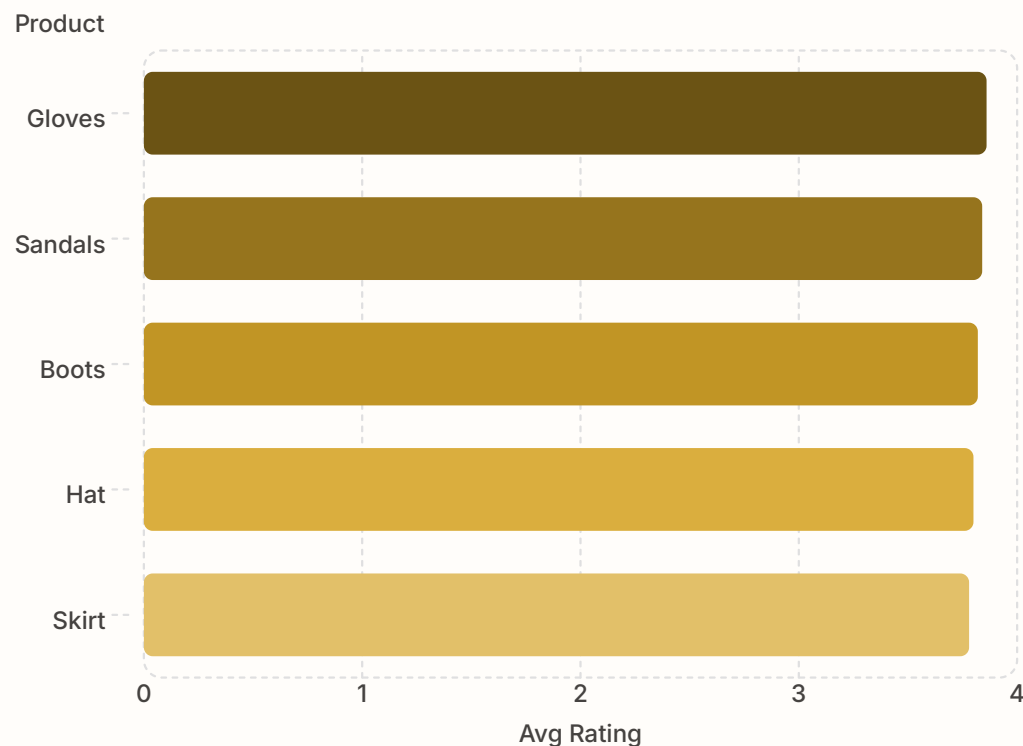
Non-Subscribers Dominate Volume

Non-subscribers (2,847 customers) generated **\$170,436** vs. subscribers' (1,053) **\$62,645**. However, average spend is nearly identical — **\$59.87** vs. **\$59.49**.

Additionally, 958 repeat buyers (5+ purchases) are subscribers vs. 2,518 non-subscribers, suggesting subscription adoption among loyal customers remains an opportunity.

Top Products & Discount Patterns

Top 5 Products by Review Rating

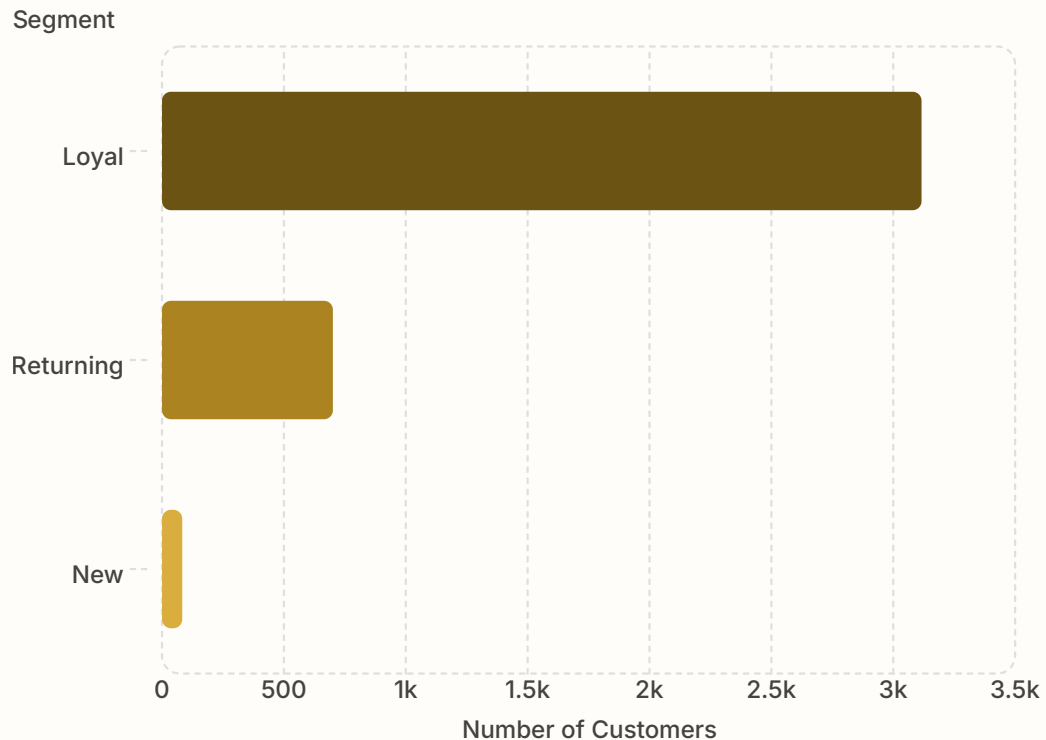


Most Discount-Dependent Products

- **Hat** — 50.00% discounted purchases
- **Sneakers** — 49.66%
- **Coat** — 49.07%
- **Sweater** — 48.17%
- **Pants** — 47.37%

839 customers used discounts yet spent above the average purchase amount.

Customer Segmentation



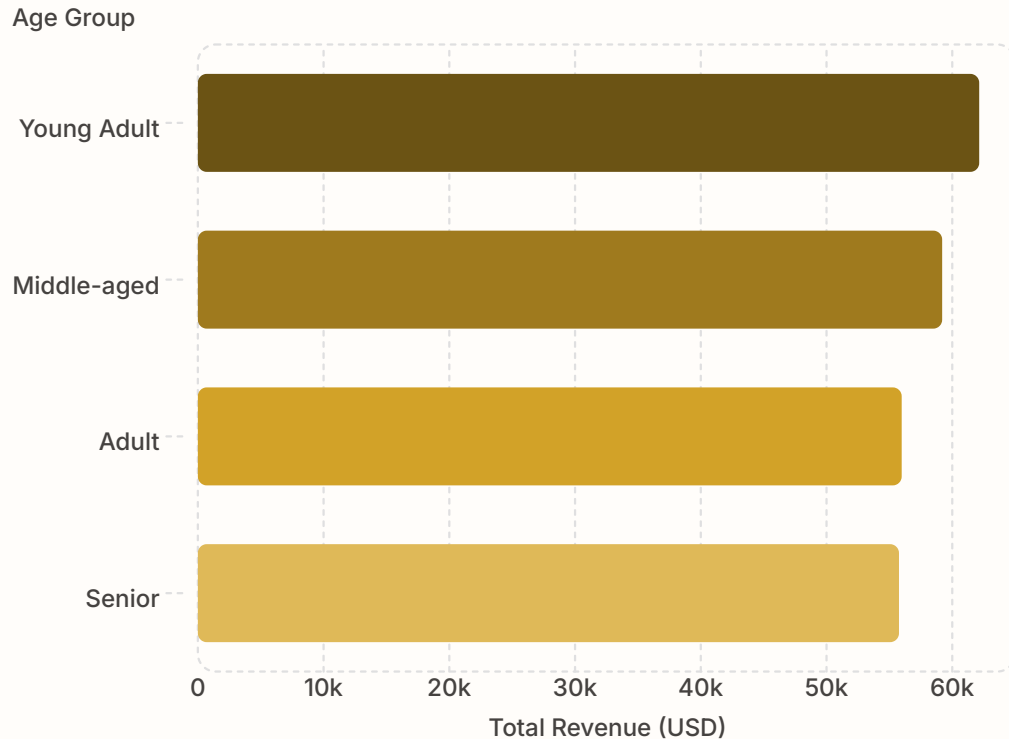
Loyal Customers Dominate

Customers were classified into three segments based on purchase history:

- **Loyal** — 3,116 customers (80% of base)
- **Returning** — 701 customers
- **New** — 83 customers

With 958 of 3,477 repeat buyers already subscribed, there is clear opportunity to convert Returning and New segments through loyalty programs.

Revenue by Age Group & Category



Young Adults Lead — But Evenly Spread

Revenue is surprisingly balanced across age groups. **Young Adults** lead at **\$62,143**, with Seniors close behind at **\$55,763** — a difference of only ~11%.

By category, **Clothing** dominates revenue (~\$100K), followed by Accessories (~\$50K), Footwear (~\$25K), and Outerwear (~\$10K).

The illustration shows a person sitting at a desk, working on a computer. The monitor displays a Power BI dashboard with the following components:

- Top Navigation Bar:** Includes the Power BI logo, a search bar, and user profile information.
- Left Sidebar:** Contains navigation links for Home, Reports, Datasets, Pages, and other features.
- Main Dashboard Area:**
 - Created Dash:** Displays two key metrics: "Pioneer Profit" at 4919.090 and "Ancora Data" at 3163.5M.
 - Performance Metrics:** Shows "DineTime" with a 303% increase and "DineTime" with a 207% increase.
 - Profit and Loss:** A bar chart showing profit and loss over time, with a trend line indicating a positive growth.
 - Customer Engagement:** A bar chart showing customer engagement over time, with a trend line indicating a positive growth.
 - Productivity Score:** A bar chart showing productivity scores over time, with a trend line indicating a positive growth.
 - Financial Summary:** Displays three key metrics: "Revenue" at 23.1M, "Profit" at 28.9M, and "Expense" at 34.4M.

The person is using a keyboard and mouse to interact with the dashboard. A water bottle is visible on the desk next to the monitor.

3.9K Customers

Total records visualized

\$59.76 Avg Purchase
Across all transactions

3.75 Avg Rating

Customer review score

27% Subscribers
73% are non-subscribers

Made with **Gamma**

Business Recommendations

Boost Subscriptions

Promote exclusive benefits to convert the 73% non-subscriber base.

Customer Loyalty Programs

Reward repeat buyers to move Returning customers into the Loyal segment.

Review Discount Policy

Balance sales boosts with margin control — 5 products show ~50% discount dependency.

Product Positioning

Highlight top-rated products (Gloves, Sandals, Boots) in campaigns.

Targeted Marketing

Focus on high-revenue age groups and express-shipping users.